

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Open-Label, Multicenter Study to Compare the Efficacy and Safety of Arlocabtagene Autoleucel (BMS-986393), a GPRC5D-directed CAR-T Cell Therapy, Versus Standard Regimens in Adult Participants With Relapsed or Refractory and Lenalidomide-exposed Multiple Myeloma **Short Title/ Acronym:** QUINTESSENTIAL-2 **Protocol Number:** CA088-1007 **NCT Number:** NCT06615479 **Sponsor Name:** Celgene Corporation (Juno Therapeutics, Inc., a Bristol-Myers Squibb Company) **Investigational Product:** Arlocabtagene autoleucel (arlo-cel / BMS-986393) - GPRC5D-directed autologous CAR T-cell therapy **Phase of Development:** Phase 3 **Medical Monitor:** GSM-CT Representative (Bristol-Myers Squibb Clinical Operations)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To compare the efficacy and safety of arlocabtagene autoleucel (arlo-cel) versus standard of care (SOC) regimens in adult participants with relapsed or refractory multiple myeloma (RRMM) who have been exposed and are refractory to lenalidomide. * **Secondary Objectives:** To evaluate overall survival (OS), overall response rate (ORR), minimal residual disease (MRD)-negative status, complete response rate, time to response, duration of response, pharmacokinetics, and patient-reported quality of life outcomes.

2.2 Background & Rationale Despite advances in multiple myeloma (MM) treatment, nearly all patients eventually relapse. MM refractory to lenalidomide—an immunomodulatory drug (IMiD) widely used in frontline and maintenance therapies—poses a significant challenge, as the disease is less likely to respond to subsequent treatments. G protein-coupled receptor class C group 5 member D (GPRC5D) is a promising therapeutic target highly expressed on malignant plasma cells with limited expression elsewhere. Arlo-cel, a GPRC5D-directed autologous CAR T-cell therapy, demonstrated a 96% ORR in patients with 1-3 prior lines of therapy in a Phase 1 study, prompting this Phase 3 confirmatory trial to evaluate its superiority against current standard regimens.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Standard Regimens: DPd [daratumumab,

pomalidomide, dexamethasone] or Kd [carfilzomib, dexamethasone])
* **Blinding:** Open-label * **Randomization:** 1:1 ratio

3.2 Planned Enrollment & Duration * **Target** **\$N\$:** 440 adult patients * **Number of Sites:** ~126 sites globally (across North America, South America, Europe, Asia, and Australia) * **Trial Status:** RECRUITING (Status Description: Currently recruiting participants) * **Estimated Study Start:** 2025 * **Estimated Primary Completion:** Follow-up for ≤ 5 years after the last patient is randomized, with a subsequent long-term follow-up study (≤ 15 years post-infusion) for patients receiving arlo-cel.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a documented diagnosis of MM as per International Myeloma Working Group Criteria. 2. Participants must have relapsed or refractory multiple myeloma (RRMM) with measurable disease during screening. 3. Received at least 1 but no greater than 3 prior MM regimens (which may include a proteasome inhibitor, an IMiD, and an anti-CD38 monoclonal antibody) and have prior exposure/refractoriness to lenalidomide. 4. Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1 and adequate organ function.

4.2 Exclusion Criteria 1. Known active or history of central nervous system (CNS) involvement of Multiple Myeloma. 2. Solitary plasmacytomas or non-secretory myeloma without other evidence of measurable disease. 3. Need for urgent treatment due to rapidly progressing MM. 4. Prior treatment with GPRC5D-targeted therapy.

4.3 Trial Phase Requirements * **Criteria Type:** PHASE_REQUIREMENT * **Phase ID:** PHASE3 * **Criteria Text:** Trial must be in PHASE3

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Arm A: Single infusion of arlo-cel at the recommended Phase 2 dose (RP2D) of $\$150 \times 10^6$ CAR T cells. * **Route of Administration:** Intravenous infusion (following leukapheresis and lymphodepleting chemotherapy). * **Duration of Treatment:** Single infusion, followed by long-term observation.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Bridging therapy with DPd or Kd within 3 days of leukapheresis. Lymphodepleting chemotherapy is administered prior to arlo-cel infusion for Arm A. Standard of Care (DPd or Kd) dosed per labeling for Arm B. * *Lymphodepleting Agent / Associated Drug:* cyclophosphamide * *Drug Preferred Name:* cyclophosphamide * *Trade Names:* Cytosan, Endoxana * *ATC Code:* L01AA01 * **Prohibited:** Other concurrent investigational therapies or previous GPRC5D targeted agents.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Eligibility, Disease Assessment, Leukapheresis (Arm A) | |
Baseline | Day 0 | Randomization, Lymphodepletion, Arlo-cel Infusion / Start of SOC Therapy | | **Interim** | Standard Oncology Cadence | Safety Labs, Efficacy & Response Assessments (IMWG criteria), MRD Evaluation, QoL | | **End of Study** | Up to 5 Years | Final Efficacy Assessment, Survival Tracking, Transition to 15-Year LTFU (Arm A) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Progression-free survival (PFS) and minimal residual disease (MRD) negativity in complete response. * **Measurement Method:** Evaluated per International Myeloma Working Group (IMWG) criteria and central MRD testing (e.g., via next-generation sequencing/flow cytometry).

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Overall Response Rate (ORR), Complete Response Rate, Time to Response, and Duration of Response * Pharmacokinetics and patient-reported Quality of Life (QoL) outcomes

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard reporting with specialized monitoring for CAR-T specific toxicities, including Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS).
 - **Serious Adverse Events (SAEs):** Routine 24-hour reporting timeline per standard pharmacovigilance protocols.
 - **Data Monitoring Committee (DMC):** Yes, an independent Data Monitoring Committee is established for this trial to evaluate safety.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy endpoints. Safety Set for all treated patients.
- **Sample Size Justification:** Approximately 440 patients, properly powered to evaluate the superiority of arlo-cel over standard regimens in improving PFS.
- **Handling of Missing Data:** Standard oncology time-to-event methodologies; censoring rules applied for PFS and OS analyses.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Regular onsite and remote monitoring for data verification, protocol adherence, and CAR-T cell handling/chain of identity compliance.
 - **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking with data clarification forms.
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11. Study Identifiers & Governance

- **Primary ID:** CA088-1007
 - **Registry Number:** NCT06615479
 - **Posting ID:** 464795020715479458
 - **Sponsor/Lead Organization (Company):** Celgene Corporation (Bristol Myers Squibb)
 - **Study Title:** QUINTESSENTIAL-2
 - **Source Level:** 5
 - **Official Regulatory Title:** A Phase 3, Randomized, Open-Label, Multicenter Study to Compare the Efficacy and Safety of Arlocabtagene Autoleucel (BMS-986393), a GPRC5D-directed CAR-T Cell Therapy, Versus Standard Regimens in Adult Participants With Relapsed or Refractory and Lenalidomide-exposed Multiple Myeloma
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12. Clinical Framework (Multiple Myeloma Specific)

- **Primary Condition (Disease Area):** Relapsed or Refractory Multiple Myeloma (RRMM)
- **Diagnosis Threshold:** IMWG Criteria for RRMM; 1-3 prior lines of therapy
- **Preferred UMLS Name / Preferred Name:** Multiple Myeloma
- **Semantic Type:** Neoplastic Process
- **Disease Definition:** A bone marrow-based plasma cell neoplasm characterized by a serum monoclonal protein and skeletal destruction with osteolytic lesions, pathological fractures, bone pain, hypercalcemia, and anemia. Clinical variants include non-secretory myeloma, smoldering myeloma, indolent myeloma, and plasma cell leukemia. (WHO, 2001)
- **MeSH Code:** D009101
- **HPO Code:** HP:0006775
- **SNOMED ID:** 109989006
- **SNOMED Hierarchy:**
 - 269475001 | Malignant neoplasm of lymphoid, hematopoietic and/or related tissue | Malignant neoplasm of lymphoid, hematopoietic and/or related tissue (disorder)

- 415111003 | Plasma cell neoplasm | Plasma cell neoplasm (disorder)
- 1162576007 | Plasma cell myeloma | Plasma cell myeloma (morphologic abnormality)
- 64572001 | Disease | Disease (disorder)
- **Optimization Target:** Lenalidomide-exposed/refractory disease management

13. Study Procedures & Methodology

- **Management Pathway:** Leukapheresis, bridging therapy, lymphodepletion, and single CAR-T infusion
- **Education Protocol (HCP):** Site certification for CAR-T administration, chain of identity processing, and CRS/ICANS adverse event management.
- **Education Protocol (Patient):** Cellular therapy education, hospitalization protocols, and long-term adverse event symptom tracking.
- **Quality Audit Mechanism:** Independent DMC safety reviews and 15-year long-term follow-up tracking for gene therapy.

14. Observation & Follow-Up Schedule

- **Total Duration:** Primary follow-up up to 5 years post-randomization; 15 years for CAR-T long-term follow-up (LTFU)
- **Onsite Visit Cadence:** Standard oncology assessment cadence (intensive post-infusion days 1-28, then standard intervals)
- **Remote Monitoring Frequency:** Standard monthly/Q4W reviews for safety and survival tracking
- **Checklist Review Interval:** Regular intervals to assess PFS, OS, and MRD negativity status

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]